

Lesson 4-3: Estimate the Percent of a Number

Grade 6 Mathematics · Standard 6.AT.4

Name: _____ Date: _____

These are MSTAR-style questions written for this curriculum to rehearse the format. They are not official Maryland assessment items. MSTAR — Maryland's new state test, first given in spring 2027 — uses the same kinds of questions you see here: selected response, select-all, two-part evidence questions, and written responses.

Part A — Skills Check

Warm up with the lesson's core skills — one best answer each.

Question 1 SELECTED RESPONSE · 1 POINT

Margaret's phone has 6 GB of storage and 4 GB is used. About what percent is used?

- ☐ A About 25%
- ☐ B About 40%
- ☐ C About 70%
- ☐ D About 150%

Question 2 SELECTED RESPONSE · 1 POINT

On the double number line for 6 GB, what amount lines up with 25%?

- ☐ A 1.25 GB
- ☐ B 1.5 GB
- ☐ C 2.5 GB
- ☐ D 4 GB

Question 3 SELECTED RESPONSE · 1 POINT

Which benchmark percents make good landmarks on a number line?

- ☐ A 0%, 25%, 50%, 75%, 100%
- ☐ B 3%, 17%, 62%, 94%
- ☐ C Only 50%
- ☐ D Only 100%

Question 4 SELECTED RESPONSE · 1 POINT

To estimate 8.6% of 216 by rounding, which numbers should you use?

- ☐ A 1% of 216
- ☐ B 8% of 300
- ☐ C 10% of 220
- ☐ D 86% of 216

Question 5 SELECTED RESPONSE · 1 POINT

What is 10% of 220?

- ☐ A 2.2
- ☐ B 22
- ☐ C 44
- ☐ D 220

Question 6 SELECTED RESPONSE · 1 POINT

The Green-Team recycled 216 pounds; 8.6% was metal. Using compatible numbers 10% and 200, the estimate is:

- ☐ A 2 pounds
- ☐ B 20 pounds
- ☐ C 86 pounds
- ☐ D 208 pounds

Part B — Test-Format Questions

These match the state test's formats: two-part questions, select-all, and written responses.

Question 7**TWO-PART QUESTION****Part A**

A jacket originally priced at \$78.99 is on sale for 25% off. Which is the best estimate of the discount amount?

Fill in the bubble next to the one best answer.

- ☐ A \$8
- ☐ B \$10
- ☐ C \$20
- ☐ D \$40

Part B

Answer Part A before Part B — Part B asks about your reasoning.

Which benchmark strategy gives the most accurate estimate in Part A?

- ☐ A Round 25% to 10% and multiply by \$78.99.
- ☐ B Round \$78.99 to \$80 and find $\frac{1}{4}$ of \$80 (half of \$80 is \$40, and half of \$40 is \$20).
- ☐ C Round \$78.99 to \$100 and subtract 25.
- ☐ D Divide \$78.99 by 25.

Question 8**SELECT ALL THAT APPLY**

Select ALL benchmark percent strategies that correctly estimate values:

Mark the box next to EVERY answer that is true. More than one is correct.

- ☐ A To find 10% of any number, divide the number by 10 (move the decimal point one place left).
- ☐ B To find 5% of a number, find 10% and take half of that value.
- ☐ C To find 50% of a number, multiply the number by 2.
- ☐ D To find 20% of a number, find 10% and double it.
- ☐ E To find 1% of a number, divide the number by 100 (move the decimal point two places left).

Part C — Show Your Thinking

Answer in writing, the way the state test's constructed responses work.

Question 9**WRITTEN RESPONSE · 2 POINTS**

Write your own estimation problem about percent, using a real situation from your life. Then solve it with a benchmark and say whether your estimate runs high or low.

Write your answer in complete sentences. Show or explain your mathematical thinking.

Sentence starters: My situation: ____ . ____% is near the benchmark ____ , so my estimate is ____ .

When you finish, check your reasoning with your teacher or family. Your teacher has the scoring guide for the written responses.